

and pleased many behavioralists. (Fama's paper, "Market Efficiency, Long-Term Returns, and Behavioral Finance," noting this concession at the Social Science Research Network, is one of the most popular investment downloads on the web site.) The *Journal* article also featured remarks by Roger Ibbotson, founder of Ibbotson Associates: "There is a shift taking place," Ibbotson observed. "People are recognizing that markets are less efficient than we thought."⁹

As Meir Statman eloquently put it, "Standard finance is the body of knowledge built on the pillars of the arbitrage principles of Miller and Modigliani, the portfolio principles of Markowitz, the capital asset pricing theory of Sharpe, Lintner, and Black, and the option-pricing theory of Black, Scholes, and Merton."¹⁰ Standard finance theory is designed to provide mathematically elegant explanations for financial questions that, when posed in real life, are often complicated by imprecise, inelegant conditions. The standard finance approach relies on a set of assumptions that oversimplify reality. For example, embedded within standard finance is the notion of *Homo economicus*, or rational economic man. It prescribes that humans make perfectly rational economic decisions at all times. Standard finance, basically, is built on rules about how investors "should" behave, rather than on principles describing how they actually behave. Behavioral finance attempts to identify and learn from the human psychological phenomena at work in financial markets and within individual investors. Behavioral finance, like standard finance, is ultimately governed by basic precepts and assumptions. However, standard finance grounds its assumptions in idealized financial behavior; behavioral finance grounds its assumptions in observed financial behavior.

Efficient Markets versus Irrational Markets

During the 1970s, the standard finance theory of market efficiency became the model of market behavior accepted by the majority of academics and a good number of professionals. The efficient market hypothesis had matured in the previous decade, stemming from the doctoral dissertation of Eugene Fama. Fama persuasively demonstrated that in a securities market populated by many well-informed investors, investments will be appropriately priced and will reflect all available information. There are three forms of the efficient market hypothesis:

1. The "Weak" form contends that all past market prices and data are fully reflected in securities prices; that is, technical analysis is of little or no value.